



Python Programming Fundamentals

Driving the Shop Class

Programming Fundamentals Team

SETU

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Outline

1. Basic Shop Driver
2. Iterating Over Products
3. Finding a Product
4. Removing a Product
5. A Simple Text Menu
6. Putting It All Together



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1. Basic Shop Driver

```
from product import Product
from shop import Shop

def main():
    my_shop = Shop("Green Grocer")

    my_shop.add_product(Product("Apple", 1.99, 101))
    my_shop.add_product(Product("Bread", 2.50, 102))
    my_shop.add_product(Product("Milk", 1.20, 103))
    my_shop.add_product(Product("Butter", 3.00, 104))

    print(my_shop)
    my_shop.display_all()
    print(f"\nTotal stock value: €{my_shop.total_value():.2f}")
```



1. Basic Shop Driver

```
if __name__ == "__main__":  
    main()
```



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2.1 Printing formatted output

```
def print_all_products(shop: Shop):
    products = shop.get_all_products()
    print(f"\n{'ID':>5} {'Name':<20} {'Price':>8}")
    print("-" * 37)
    for p in products:
        print(f"{p.get_product_id():>5} "
              f"{p.get_name():<20} "
              f"€{p.get_price():>7.2f}")
    print("-" * 37)
    print(f"{'Total':>27} €{shop.total_value():>7.2f}")
```

Output:

ID	Name	Price



2.1 Printing formatted output

101	Apple	€	1.99
102	Bread	€	2.50



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3. Finding a Product

```
def find_product(shop: Shop):  
    name = input("Enter product name to find: ")  
    result = shop.find_by_name(name)  
    if result:  
        print(f"Found: {result}")  
    else:  
        print(f"No product named '{name}' found.")
```



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4. Removing a Product

```
def remove_product(shop: Shop):  
    name = input("Enter product name to remove: ")  
    if shop.remove_product(name):  
        print(f"'{name}' removed successfully.")  
    else:  
        print(f"'{name}' not found – nothing removed.")
```



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5. A Simple Text Menu

```
def show_menu():
    print("\n=== Shop Menu ===")
    print("1. List all products")
    print("2. Find product")
    print("3. Remove product")
    print("4. Show total value")
    print("0. Exit")

def main():
    shop = Shop("Green Grocer")
    # ... add initial products ...
    running = True
    while running:
        show_menu()
        choice = input("Choose: ")
        if choice == "1":    shop.display_all()
```



5. A Simple Text Menu

```
elif choice == "2": find_product(shop)
elif choice == "3": remove_product(shop)
elif choice == "4": print(f"€{shop.total_value():.2f}")
elif choice == "0": running = False
```



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6. Putting It All Together

File	Responsibility
<code>product.py</code>	Product class — data + validation
<code>shop.py</code>	Shop class — manages list of Products
<code>driver.py</code>	<code>main()</code> — menu + user interaction

This three-file structure is a mini **three-layer architecture**:

- Model (Product, Shop)
- Controller (driver functions)
- View (print functions)

Thanks for Watching - Any questions?

